

1 - PURPOSE

Nicotinic acid (niacin) is a micronized API with poor flow properties making the production of high dose sustained release tablets by direct compression a real challenge. This study aims to evaluate wet granulation as a simple and efficient process to obtain high dose niacin sustained release lipid matrix tablets.

2 - INTRODUCTION

Micronized drugs have poor flow properties which make the production of high dose tablets by direct compression difficult due to homogeneity, dust generation and process variability. The drug developers usual recourse is to formulate such compounds by a wet granulation process.

Compritol® 888 ATO (glyceryl dibehenate NF) is frequently used in sustained release systems. It is a triacylglyceride (glycerol esterified with behenic acid), with a HLB of 1 and a melting point of 70°C. Insoluble and lipophilic, it forms a release retarding matrix which entraps drug and slowly releases it by diffusion over several hours. Being insoluble in aqueous fluids, Compritol® 888 is well-suited for wet granulation in contrast to hydrophilic polymers that may swell causing the formation of lumps [1]. The study describes the development of sustained-release niacin tablets with Compritol® 888 using a wet granulation process.

3 - EXPERIMENTAL METHODS

Niacin properties have been determined using a binocular loop and a flow tester. Niacin was blended with Compritol® 888 ATO at varying ratios. A mortar and pestle were used to granulate the powders with the addition of water. The impact of a binder (Kollidon® 30, PVP, BASF) in the powder blend was evaluated. Granules were sieved, dried overnight at 40°C in a dry oven and compressed into flat faced tablets (Korsch EK0, 14kN, 16mm/12mm), using 0.5% magnesium stearate as lubricant. The drug dose was 500 or 1000mg. Drug release studies were conducted in a USP II paddle apparatus (900mL 0.1N HCl, 37°C, 100rpm) and the dissolution profiles compared by application of the similarity factor f_2 .

4 - RESULTS AND DISCUSSION

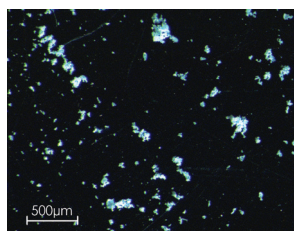


Figure 1: Macroscopic image of micronized niacin.

Micronized niacin has a particle size of 34µm (Figure 1) and exhibits very poor powder flow and compressibility (Table 1). Powder-flow is improved by granulating niacin with Compritol® 888 ATO using water as binder. The resulting granules showed good flow properties and were easily compressed into tablets. Drug release was sustained over at least 12h depending on the niacin:Compritol® ratio (Figure 2). Importantly, drug release from 90:10 blends was slowed down by simply changing the tablet dimension due to the longer drug diffusion pathway (Figure 3). However, granule cohesion was limited and tablets therefore tended to stick to the punch.

Table 1: Flow behavior of micronized niacin.

Niacin properties	100g
Flowability, s	n.d.
Angle of repose, °	>60
Hausner ratio	1.76
Carr's Index, %	43

n.d. = not detectable

Table 2: Composition of tablets prepared by wet granulation.

Ingredients	%w/w
Nicotinic acid	0-91.7
Compritol® 888 ATO	q.s.
Kollidon® 30 (if indicated)	3
Mg Stearate	0.5

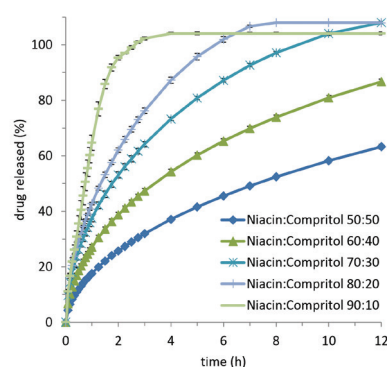


Figure 2: Drug release from tablets containing granules with various niacin/Compritol® ratios (dose 500mg, no binder).

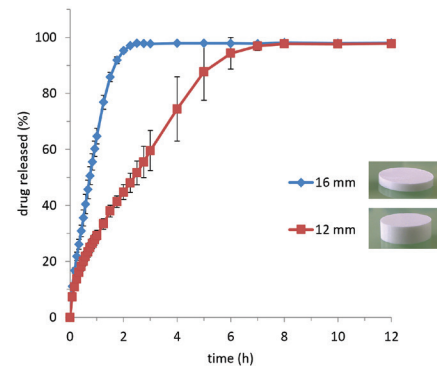


Figure 3: Impact of tablet dimension on drug release (dose 500 mg, drug:lipid ratio 90:10).

The addition of 3% of PVP yielded tablets with an approximately two-fold increase in tensile strength (Figure 4). Adhesion to the punch was prevented, resulting in tablets with a smooth surface. The increase in tablet tensile strength slightly reduced drug release over time but not significantly (Figure 5). The niacin:Compritol® 888 ratio affected tablet size with higher drug loads producing smaller tablets (Figure 6). This is of particular interest since higher drug loads generally produce large cumbersome tablets.

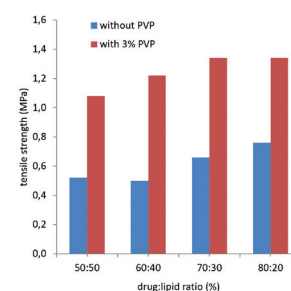


Figure 4: Effect of binder addition on tablet tensile strength.

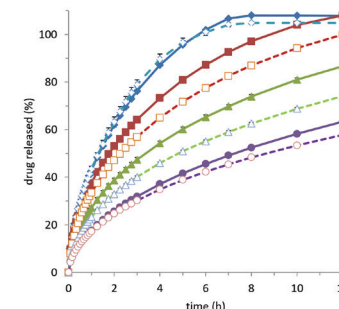


Figure 5: Influence of PVP on niacin release: filled symbols without PVP, open symbols with PVP (dose 500mg).

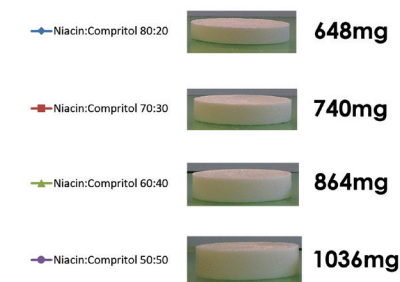


Figure 6: Effect of the drug:lipid ratio on tablet height and weight (dose 500mg).

1000mg niacin tablets were prepared varying the drug:lipid ratios and the results indicate that drug release can be effectively sustained using 10% of Compritol® 888 (Figure 7). Drug release kinetics similar to the marketed product Niaspan® were obtained ($f_2 > 50$). The low Compritol® 888 concentration enables a reasonable tablet weight to be maintained despite high drug loading (Figure 8). Again, tablet height was affected by the drug:lipid ratio (Figure 8).

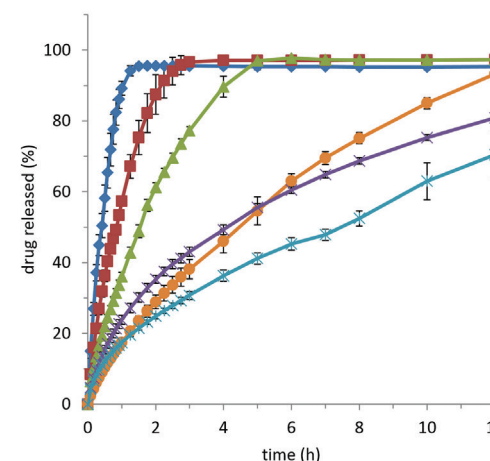


Figure 7: Drug release from tablets containing different niacin/Compritol® ratios (dose 1000mg).

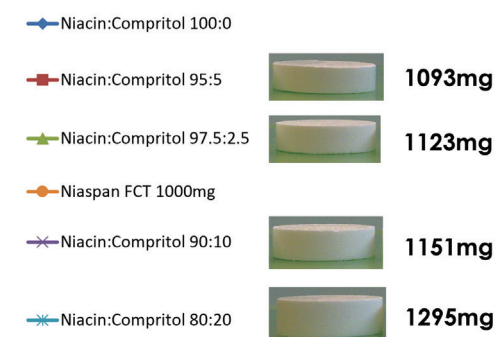


Figure 8: Effect of the drug:lipid ratio on tablet size and weight (dose 1000mg).

5 - CONCLUSION

Being a micronized and freely soluble drug, sustained release niacin tablets can be challenging to formulate. The formulation of a lipid matrix tablet by simple wet granulation-compression is a viable approach. The addition of PVP is shown to improve the quality of tablets with limited effect on drug release. High dose lipid matrix tablets delivered sustained release comparable to an existing authorized medicine. The ratio of drug to excipient enables tablet height and size to be considered as part of the overall dosage form design.

6 - REFERENCES

- [1] Jeon, I. et al. J Pharm Bioallied Sci. 2011 Apr-Jun; 3(2): 213-220.